

General Risk Assessment Form

Date: (1) 2026.02.24	Assessed by: (2) Zheng Xu	Checked / Validated* by: (3)	Location: (4)	Assessment ref no (5)	Review date: (6)
Task / premises: (7) • Development and evaluation of a Brain-Computer Interface (BCI) system using Reinforcement Learning (RL) to control a low-cost robotic arm, based on public EEG datasets. • Activities include: Software development (Python, RL libraries, simulation environments like PyBullet/Gazebo), EEG data processing (using existing datasets), robotic arm setup and operation (physical low-cost arm), and system testing (primarily simulation, potentially limited physical testing).					
Activity (8)	Hazard (9)	Who might be harmed and how (10)	Existing measures to control risk (11)	Risk rating (12)	Result (13)
Prolonged computer uses for software development	Incorrect or prolonged use of Display Screen Equipment (DSE)	Project student. Potential for musculoskeletal issues (e.g., RSI, back/neck pain) from prolonged poor posture or repetitive actions, and visual fatigue or headaches from extended screen time.	1. University DSE guidelines are followed. 2. Regular breaks taken from the screen (e.g., 5-10 mins per hour). 3. Workstation (chair, monitor, keyboard) is set up ergonomically to maintain good posture.	Severity: Low Likelihood: Medium Score: Low	A (Adequately Controlled)

Result : T = trivial, A = adequately controlled, N = not adequately controlled, action required, U = unknown risk

Activity (8)	Hazard (9)	Who might be harmed and how (10)	Existing measures to control risk (11)	Risk rating (12)	Result (13)
Use of general electrical equipment (PC, Monitor etc.)	Contact with live parts or faulty equipment	Project student. Potential for electric shock from contact with live parts, or burns/fire from overheating/short-circuiting equipment.	1. All equipment (PC, monitor, etc.) is CE marked commercial product. 2. A visual inspection of cables and plugs is carried out before use. 3. Sockets are not overloaded. 4. Liquids are kept away from equipment.	Severity: Medium Likelihood: Low Score: Low	A (Adequately Controlled)
Lone Working	Sudden illness or accident while working alone with no immediate assistance available	Project student. A delay in getting assistance in the event of a sudden illness or accident (e.g., slip, trip) could exacerbate an injury.	1. Mobile phone kept accessible for emergencies. 2. Regular contact maintained with family/friends regarding work schedule. 3. Aware of University Security emergency number (69966) and standard emergency services (999).	Severity: High Likelihood: Low Score: Medium	A (Adequately Controlled)
General activities in the workspace	Fire caused by electrical faults, overheating equipment, or other ignition sources	Project student and others in the vicinity. Harm from burns, smoke inhalation, or injuries sustained during an emergency evacuation.	1. Familiar with fire evacuation routes and emergency procedures for the work area. 2. Workspace is kept tidy and clear of combustible materials. 3. Safe electrical practices are followed to minimise the risk of electrical fire.	Severity: High Likelihood: Low Score: Medium	A (Adequately Controlled)
Software Development & Data Management	Data loss or corruption due to software bugs, system crashes, or hardware (e.g., hard drive) failure.	Project student. Loss of hours or weeks of work (code, data, report), leading to significant project delays and stress.	1. Daily code commits to a version control system (e.g., Git/GitHub). 2. Regular backups of all project files (report, data, simulation results) to at least two separate locations (e.g., cloud storage and a physical external hard drive).	Severity: High Likelihood: Low Score: Medium	A (Adequately Controlled)

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Robotic Arm Electrical Setup & Use	Electricity: Contact with robot power supply, faulty wiring, short circuits	Project student. Electric shock, burns, fire risk from robot power system.	1. Use only the manufacturer-supplied or recommended power adapter. 2. Visually inspect robot cables and power supply before each use. 3. Ensure proper connections and avoid damaging cables. 4. Keep liquids away from the robot and its power supply. 5. Ensure workspace has functional RCD protection if possible.	Severity: Medium Likelihood: Low Score: Low	A (Adequately Controlled)
Robotic Arm Setup and Operation	Mechanical Hazards: Moving parts, unexpected movements, collision, pinching	Project student. Physical injury (bruising, pinching, impact) from contact with the moving arm. Damage to the robot or surrounding objects.	1. Define and maintain a clear, unobstructed operating zone around the robot. 2. Operate the robot at low speed settings during testing and development. 3. Implement software-based safety limits (workspace boundaries, joint angle limits, velocity/acceleration limits) referenced from baseline project {bci_baseline}. 4. Always have a physical emergency stop method readily available (e.g., power switch). 5. Avoid placing hands or objects in the robot's path while operational. 6. Perform initial tests and RL training primarily in a simulated environment.	Severity: Medium Likelihood: Low Score: Low	A (Adequately Controlled)
Physical Robotic Arm Control via BCI	Impact injury from robotic arm movement: The arm may strike a person standing nearby during	Project student and others in the vicinity. Potential for bruising, cuts, or more serious injury if struck by the moving arm, particularly	1. Clear warning given before starting arm operation. 2. Maintain a minimum 0.5m safety distance from the arm during operation. 3. Operate at reduced speed (velocity limited to 80 ticks/s in software). 4. Use software-defined soft joint limits to prevent extreme movements. 5. Emergency stop (power switch) readily	Severity: Medium Likelihood: Low Score: Low	A (Adequately Controlled)

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	operation, especially during unexpected movements or control errors.	at joints or end-effector.	accessible. 6. Supervision required during physical testing.		
Physical Robotic Arm Control via BCI	Object displacement or damage: The robotic arm may knock over objects on the work surface (equipment, drinks, tools) causing damage or creating secondary hazards.	Project student and equipment. Potential for damage to electronic equipment if struck, liquid spills creating electrical hazard, or falling objects causing injury.	1. Clear the workspace of unnecessary objects before arm operation. 2. Secure essential items (laptop, monitor) at safe distance. 3. No liquids allowed within arm's reach during operation. 4. Implement workspace boundaries in software to constrain arm movement. 5. Use auto-recenter feature to prevent drift toward workspace edges.	Severity: Low Likelihood: Medium Score: Low	A (Adequately Controlled)
Physical Robotic Arm Control via BCI	Pinch/crush injury: Fingers or clothing may be caught	Project student. Potential for pinch injuries to fingers, or entanglement of	1. Keep hands and loose clothing away from joints during operation. 2. Tie back long hair when working near the arm. 3. Stop arm operation before making any adjustments.	Severity: Medium Likelihood: Low Score: Low	A (Adequately Controlled)

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	between moving joints of the robotic arm during operation or maintenance.	loose clothing/hair in moving parts.	4. Movement velocity limited in software for slower, predictable motion. 5. Visual monitoring of arm position at all times during operation.		
Physical Robotic Arm Control via BCI	Repetitive unexpected arm movements: BCI classification errors may cause the arm to move in unintended directions repeatedly, creating unpredictable behavior.	Project student and others nearby. Confusion about arm movement intent, potential for surprise contact or psychological stress from erratic behavior.	1. RL model limits action space to 4 directions (left/right/up/down). 2. Maximum step count (30-50) prevents infinite loops. 3. Auto-stop when target is reached or maximum steps exceeded. 4. Visual indicator showing predicted intent before movement. 5. Physical emergency stop always accessible.	Severity: Low Likelihood: Medium Score: Low	A (Adequately Controlled)

Action plan (14)

Ref No	Further action required	Action by whom	Action by when	Done
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1	Continue to follow DSE guidelines and review workstation setup if any discomfort occurs.	Zheng Xu	Ongoing	
2	When working off-campus, ensure the environment is safe and inform family/flatmates of the work schedule.	Zheng Xu	Ongoing	
3	Periodically note and remain familiar with emergency exits and evacuation routes in the working area.	Zheng Xu	Ongoing	
4	Strictly adhere to the daily/weekly backup and version control schedule throughout the project lifecycle.	Zheng Xu	Ongoing	
5	Implement and test software safety limits (workspace, velocity, joint angles) for the robotic arm in simulation.	Zheng Xu	Before physical tests	
6	Define and physically mark the safe operating zone around the robotic arm before any physical operation.	Zheng Xu	Before physical tests	
7	Identify and test the physical emergency stop procedure (e.g., power switch location and accessibility).	Zheng Xu	Before physical tests	
8	Establish and communicate verbal warning protocol before initiating robotic arm movement in physical tests.	Zheng Xu	Before physical tests	
9	Clear and secure workspace before each physical arm control session. Remove loose items and liquids from arm's reach.	Zheng Xu	Before each session	
10	Brief any observers on safety procedures and maintain minimum 0.5m distance during arm operation.	Zheng Xu	Ongoing	

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Notes to accompany General Risk Assessment Form

This form is the one recommended by Health & Safety Services, and used on the University's risk assessment training courses. It is strongly suggested that you use it for all new assessments, and when existing assessments are being substantially revised. However, its use is not compulsory. Providing the assessor addresses the same issues; alternative layouts may be used.

- (1) **Date** : Insert date that assessment form is completed. The assessment must be valid on that day, and subsequent days, unless circumstances change and amendments are necessary.
- (2) **Assessed by** : Insert the name and signature of the assessor. For assessments other than very simple ones, the assessor should have attended the University course on risk assessments (link to STDU)
- (3) **Checked / Validated* by** : delete one.

Checked by : Insert the name and signature of someone in a position to check that the assessment has been carried out by a competent person who can identify hazards and assess risk, and that the control measures are reasonable and in place. The checker will normally be a line manager, supervisor, principal investigator, etc. Checking will be appropriate for most risk assessments.

Validated by : Use this for higher risk scenarios, eg where complex calculations have to be validated by another "independent" person who is competent to do so, or where the control measure is a strict permit-to-work procedure requiring thorough preparation of a workplace. The validator should also have attended the University's risk assessment course or equivalent, and will probably be a chartered engineer or professional with expertise in the task being considered. Examples of where validation is required include designs for pressure vessels, load-bearing equipment, lifting equipment carrying personnel or items over populated areas, and similar situations.

- (4) **Location** : insert details of the exact location, ie building, floor, room or laboratory etc
- (5) **Assessment ref no** : use this to insert any local tracking references used by the school or administrative directorate
- (6) **Review date** : insert details of when the assessment will be reviewed as a matter of routine. This might be in 1 year's time, at the end of a short programme of work, or longer period if risks are known to be stable. Note that any assessment must be reviewed if there are any significant changes – to the work activity, the vicinity, the people exposed to the risk, etc
- (7) **Task / premises** : insert a brief summary of the task, eg typical office activities such as filing, DSE work, lifting and moving small objects, use of misc electrical equipment. Or, research project [title] involving the use of typical laboratory hardware, including fume cupboards, hot plates, ovens, analysis equipment, flammable solvents, etc.
- (8) **Activity** : use the column to describe each separate activity covered by the assessment. The number of rows is unlimited, although how many are used for one assessment will depend on how the task / premises is sub-divided. For laboratory work, activities in one particular lab or for one particular project might include; use of gas cylinders, use of fume cupboard, use of computer or other electrical equipment, use of lab ovens, hot plates or heaters, use of substances hazardous to health, etc
- (9) **Hazard** : for each activity, list the hazards. Remember to look at hazards that are not immediately obvious. For example, use of a lathe will require identification of the machine hazards, but also identification of hazards associated with the use of cutting oils (dermatitis), poor lighting, slipping on oil leaks, etc. The same activity might well have several hazards associated with it. Assessment of simple chemical risks (eg use of cleaning chemicals in accordance with the instructions on the bottle) may be recorded here. More complex COSHH

assessments eg for laboratory processes, should be recorded on the specific COSHH forms (link).

- (10) **Who might be harmed and how** : insert everyone who might be affected by the activity and specify groups particularly at risk. Remember those who are not immediately involved in the work, including cleaners, young persons on work experience, maintenance contractors, Estates personnel carrying out routine maintenance and other work. Remember also that the risks for different groups will vary. Eg someone who needs to repair a laser may need to expose the beam path more than users of the laser would do. Vulnerable groups could include children on organised visits, someone who is pregnant, or employees and students with known disabilities or health conditions (this is not a definitive list).

For each group, describe how harm might come about, eg an obstruction or wet patch on an exit route is a hazard that might cause a trip and fall; use of electrical equipment might give rise to a risk of electric shock; use of a ultraviolet light source could burn eyes or skin.

- (11) **Existing measures to control the risk** : list all measures that already mitigate the risk. Many of these will have been implemented for other reasons, but should nevertheless be recognised as means of controlling risk. For example, restricting access to laboratories or machine rooms for security reasons also controls the risk of unauthorised and unskilled access to dangerous equipment. A standard operating procedure or local rules (eg for work with ionising radiation, lasers or biological hazards) will often address risks. Some specific hazards may require detailed assessments in accordance with specific legislation (eg COSHH, DSEAR, manual handling, DSE work). Where this is the case, and a detailed assessment has already been done in another format, the master risk assessment can simply cross-reference to other documentation. For example, the activity might be use of a carcinogen, the hazard might be exposure to hazardous substances, the existing control measures might all be listed in a COSHH assessment. Controls might also include use of qualified and/or experienced staff who are competent to carry out certain tasks; an action plan might include training requirements for other people who will be carrying out those tasks.

- (12) **Risk Rating** : the simplest form of risk assessment is to rate the remaining risk as high, medium or low, depending on how likely the activity is to cause harm and how serious that harm might be.

The risk is **LOW** - if it is most unlikely that harm would arise under the controlled conditions listed, and even if exposure occurred, the injury would be relatively slight.

The risk is **MEDIUM** - if it is more likely that harm might actually occur and the outcome could be more serious (eg some time off work, or a minor physical injury).

The risk is **HIGH** - if injury is likely to arise (eg there have been previous incidents, the situation looks like an accident waiting to happen) and that injury might be serious (broken bones, trip to the hospital, loss of consciousness), or even a fatality.

Schools or administrative directorates may choose to use other rating systems. Typical amongst these are matrices (of 3x3, 4x4, 5x5 or even more complex) which require the assessor to select a numerical rating for both "likelihood that harm will arise" and "severity of that harm". These may give a spurious sense of accuracy and reliability – none are based on quantitative methods. There are methods of estimating risk quantitatively, and these may be appropriate for complex design of load bearing structures and the like. Advice on methods of risk assessment is available from HSS. Whatever system of assessment is adopted, it is **essential** that the assessor has received suitable training and is familiar with the meaning of the terms (or numbers) used.

- (13) **Result** : this stage of assessment is often overlooked, but is probably the most important. Assigning a number or rating to a risk does not mean that the risk is necessarily adequately controlled. The options for this column are:

T = trivial risk. Use for very low risk activities to show that you have correctly identified a hazard, but that in the particular circumstances, the risk is insignificant.

A = adequately controlled, no further action necessary. If your control measures lead you to conclude that the risk is low, and that all legislative requirements have been met (and University policies complied with), then insert A in this column.

N = not adequately controlled, actions required. Sometimes, particularly when setting up new procedures or adapting existing processes, the risk assessment might identify that the risk is high or medium when it is capable of being reduced by methods that are reasonably practicable. In these cases, an action plan is required. The plan should list the actions necessary, who they are to be carried out by, a date for completing the actions, and a signature box for the assessor to sign off that the action(s) has been satisfactorily completed. Some action plans will be complex documents; others may be one or two actions that can be completed with a short timescale.

U = unable to decide. Further information required. Use this designation if the assessor is unable to complete any of the boxes, for any reason. Sometimes, additional information can be obtained readily (eg from equipment or chemicals suppliers, specialist University advisors) but sometimes detailed and prolonged enquiries might be required. Eg is someone is moving a research programme from a research establishment overseas where health and safety legislation is very different from that in the UK.

For T and A results, the assessment is complete.

For N or U results, more work is required before the assessment can be signed off.

- (14) **Action Plan.** Include details of any actions necessary in order to meet the requirements of the information in Section 11 'Existing measures to control the risk'. Identify someone who will be responsible for ensuring the action is taken and the date by which this should be completed. Put the date when the action has been completed in the final column.